Detecting AI-Generated Text Using Lightweight Linguistic Models

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Introduction

Large language models (LLMs) such as ChatGPT have made it increasingly difficult to distinguish between human-written and AI-generated text.

This project evaluates whether **simple, interpretable models** can reliably detect AI-generated content.

Research Questions:

- Can lightweight models distinguish AI vs human text?
- What linguistic patterns differentiate them?
- How well does performance generalize?

Accuracy by Source

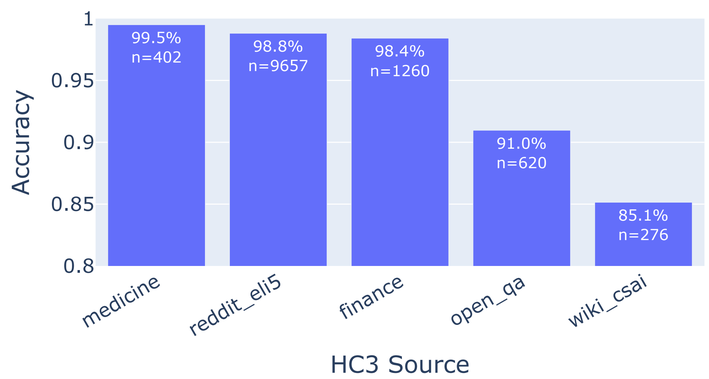

Figure 1. High accuracy across domains shows strong generalization.

Methodology

Dataset:

- HC3 (Human vs ChatGPT)
- Multiple domains (Reddit, StackExchange)

Model:

- TF-IDF features
- Logistic Regression
- Interpretable coefficients

Evaluation:

- Accuracy
- Source generalization

Model Confidence Distribution

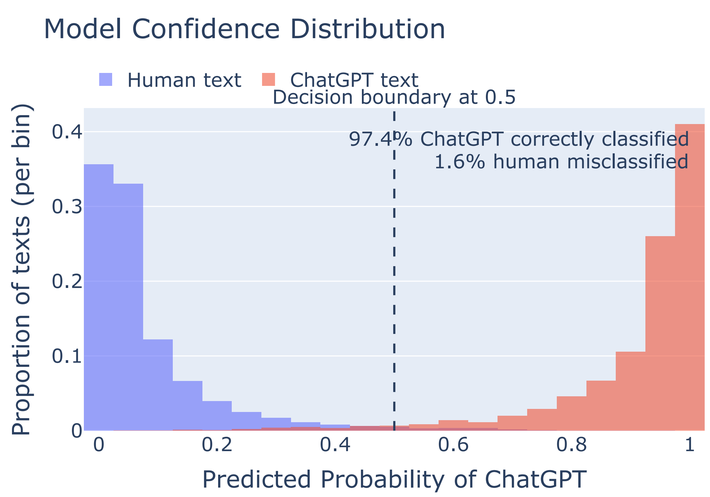

Figure 2. Strong separation between human and AI predictions.

Feature Importance

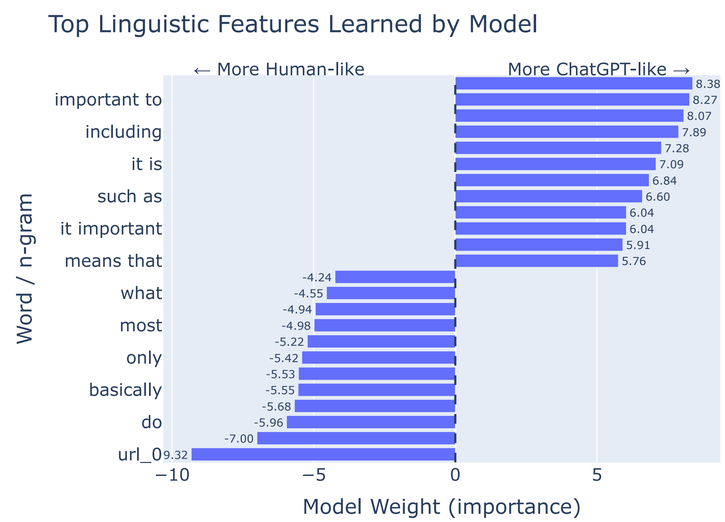

Figure 3. Linguistic features distinguishing AI vs human writing.

Findings

Observations:

- AI text is more stylistically consistent
- Specific phrases strongly indicate LLM output
- Human text is more varied

Implications:

- Lightweight models are highly effective
- Interpretability provides clear insight
- Detection may become harder as models improve

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